Onkar Jadhav

AI / Machine Learning Engineer
Pune, Maharashtra
B.Tech Computer Science, PCCOE (9.08 CGPA)

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Summary

Final-year Computer Science student (9.08 CGPA) with hands-on machine learning experience across deep learning, feature engineering, model evaluation, and AI safety. Designed StockXpert, a hierarchical deep learning system using ResNet, GRU, Bi-LSTM, and multi-head attention across 100 Nifty NSE equities, improving short-horizon return correlation by 12%. Built GovBench, an open-source LLM governance benchmark with controlled bias testing, adversarial evaluation, and hallucination detection across US, India, and EU legal systems. Also ship Stride to production as its sole engineer. Proficient in Python, TensorFlow, SQL, FastAPI, and LLM evaluation pipelines.

Education

•Bachelor of Technology in Computer Science *Nov 2022 – Jun 2026*
Pimpri Chinchwad College of Engineering, Savitribai Phule Pune University CGPA: 9.08 (up to 7th semester)
– Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming

•Higher Secondary Certificate *Jun 2022*
Shri Mhalsakant Junior College, Maharashtra State Board Score: 89.17%

Experience

•Stride — AI-First Adaptive Wellness Platform *April 2026 – Present*
Founder & Lead Engineer Remote
– Architected and shipped Stride to production as sole engineer; running live with an early private user cohort actively driving iterative product feedback ahead of public beta.
– Architected a six-layer AI-first system with multi-modal input capture, behavioral memory, natural-language understanding, five specialised domain agents, analytics, and adaptive UX.
– Separated LLM understanding from a deterministic four-component calculator (BMR, NEAT, EAT, TEF) to eliminate opaque LLM arithmetic errors while retaining natural-language input.
– Built natural-language workout parsing that converts free text into structured sets, reps, duration, and MET values without rigid input templates.
– Selected Convex for real-time reactive state over REST + WebSockets and OpenRouter for multi-provider LLM routing across five concurrent agent types.

Engineering Projects

•StockXpert — *AI-Powered Deep Learning Equity Prediction Platform* *Jan 2026 – May 2026*
Python, TensorFlow, FastAPI, Next.js, TypeScript, Cloudflare R2, AWS
– Architected a hierarchical deep learning system combining ResNet, GRU, Bi-LSTM, and multi-head attention to model cross-horizon price dependencies across all **100 Nifty NSE equities**.
– Constructed a leakage-free feature pipeline from raw OHLCV data with RSI, MACD, ADX, Bollinger Bands, SMA/EMA distance, and volatility indicators using strict temporal train/validation/test splits.
– Designed a multi-objective Huber loss with directional regularisation and tail-event oversampling, achieving a **12% improvement in short-horizon return correlation** over baseline.
– Built a snapshot-first serving layer with market-aware TTL caching, delivering **sub-second API latency** across all 100 Nifty stocks during market hours.

•GovBench — *LLM Governance Evaluation* *Mar 2026 – Present*
Python, LLM Evaluation Pipelines, Hallucination Detection, Adversarial Testing, AI Safety
– Designed and open-sourced an LLM evaluation framework assessing model readiness for judicial and governmental deployment across 6 pillars including demographic bias, transparency, and minority protection.
– Implemented controlled demographic isolation testing across **12 identity variants** per scenario with naturalistic prompting across **US, India, and EU** legal systems.
– Evaluated 4 production LLMs across baseline, pressure, and adversarial stress modes, revealing concrete failures in minority protection and jurisdictional awareness.
– Built an automated scoring pipeline combining sentiment variance analysis, position drift tracking, and hallucination detection to emit composite deployment-readiness grades.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Java
AI / ML: TensorFlow, RAG Pipelines, pgvector, BM25, Reciprocal Rank Fusion, Agentic Workflows, Behavioral Memory, Multi-modal AI, XGBoost, OpenRouter
ML Evaluation: LLM Evaluation Pipelines, Sentiment Analysis, Hallucination Detection, Adversarial Testing, AI Safety
Backend: FastAPI, Node.js, Express.js, BullMQ, Celery, Redis, REST APIs, WebSockets
Databases: PostgreSQL (+ pgvector), MongoDB, Convex
Cloud & DevOps: AWS, Cloudflare R2, Git, GitHub